

# Power of Plenty

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## 1 Abstract

This study investigates the practical limits of portfolio diversification, the strategy of adding more assets to reduce the individual asset risk. The study simulated return distributions through a Monte Carlo simulation using a multivariate t distribution (MVT) model with 1000 iterations for different portfolio sizes ( $n$ ), with data from the S&P 500, with data spanning the years 2015 to 2025. The study evaluated diversification by studying the rate at which the individual asset risk from the simulation, often called unsystematic risk, converged towards the systematic risk, the risk that cannot be eliminated through increased assets. Using a controlled data-generating process with predefined correlation parameters, the study could, with financial theory, confirm that the model gave the expected results. The study could also from the simulation, identify a threshold at  $n = 25$ , where 95% of the unsystematic risk was eliminated, where adding more assets became negligible. The found asset threshold was then evaluated empirically for the same original dataset during the time period 2015 to 2025. The study concluded that while diversification with 25 assets was highly effective during normal market conditions, it ultimately failed during financial instability and market-wide movements where the volatility and the correlation of the assets spiked.

# Contents

|          |                                                                                 |           |
|----------|---------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Abstract</b>                                                                 | <b>1</b>  |
| <b>2</b> | <b>Introduction</b>                                                             | <b>3</b>  |
| 2.1      | Background . . . . .                                                            | 3         |
| 2.2      | Problem Discussion . . . . .                                                    | 3         |
| 2.3      | Overview of planned analysis . . . . .                                          | 4         |
| <b>3</b> | <b>Data</b>                                                                     | <b>4</b>  |
| 3.1      | Dataset . . . . .                                                               | 4         |
| 3.2      | Variables . . . . .                                                             | 5         |
| 3.3      | Limitations . . . . .                                                           | 5         |
| 3.4      | Exploratory analysis . . . . .                                                  | 6         |
| <b>4</b> | <b>Method</b>                                                                   | <b>8</b>  |
| 4.1      | Investigated relationships . . . . .                                            | 8         |
| 4.2      | Multivariate t-distribution . . . . .                                           | 8         |
| 4.3      | ECME . . . . .                                                                  | 9         |
| 4.4      | Monte Carlo simulation . . . . .                                                | 10        |
| 4.5      | Definition of Portfolio and Risk . . . . .                                      | 10        |
| 4.6      | Convergence threshold . . . . .                                                 | 11        |
| 4.7      | Model evaluation . . . . .                                                      | 12        |
| <b>5</b> | <b>Implementation</b>                                                           | <b>13</b> |
| 5.1      | Package and main function . . . . .                                             | 13        |
| 5.2      | Simulation Design for Package Validation . . . . .                              | 14        |
| 5.2.1    | Simulation Procedure . . . . .                                                  | 15        |
| 5.2.2    | Theoretical Benchmarks . . . . .                                                | 16        |
| 5.3      | Simulation results . . . . .                                                    | 16        |
| <b>6</b> | <b>Results</b>                                                                  | <b>18</b> |
| 6.1      | Variability of the standard deviation estimate . . . . .                        | 18        |
| 6.2      | Convergence of the annualized volatility towards the systematic floor . . . . . | 19        |
| <b>7</b> | <b>Conclusion</b>                                                               | <b>22</b> |
|          | <b>References</b>                                                               | <b>24</b> |

## 2 Introduction

### 2.1 Background

For most investors, the main goal is not only to maximize returns but to have a portfolio that is also predictable. Most people depend on that their money will be there when they need it, whether it is college funds or retirement money. For example, as cited in the foundational framework of behavioral finance, Prospect Theory, "*the aggravation that one experiences in losing a sum of money appears to be greater than the pleasure associated with gaining the same amount*" (Kahneman and Tversky, 1979). Because of this, for most people, the volatility of the portfolio is just as important as the return. The goal is not only to maximize returns, but also to ensure that the money does not vanish and that the portfolio can consistently generate stable returns.

The most common way to create this stability is through diversification. One of the most famous studies concerning this is "Modern Portfolio Theory", introduced by Harry Markowitz in 1952. A big takeaway in this is the idea that an investor could reduce the overall standard deviation of the portfolio by having more assets less correlated to each other. By adding more stocks that are less correlated to each other, an investor could diminish the firm-specific, unsystematic risk. Modern Portfolio Theory showed that through diversification, an increased portfolio size could slowly reduce this unsystematic risk, diminishing it, eventually down towards the systematic risk (market risk), which is the non-diversifiable component of the total portfolio risk. (Markowitz, 1952).

### 2.2 Problem Discussion

While modern portfolio theory suggests that diversification is an important tool for risk reduction, the following question will be further evaluated in this analysis:

- **The rate of the systematic risk convergence:** At what rate does the portfolio standard deviation converge towards the systematic floor, and how is the diversification benefit affected by the market structure during the chosen time period?

The study aims to analyze the concept of diversification through a simulation study with stocks from the S&P 500 during the period 2015 to 2025. The goal of this study is to study the diversification rate through the simulation, along with finding a threshold in the data where adding more assets becomes negligible. By empirically testing the threshold portfolio size, it can be evaluated how the stability and benefit of the diversification hold up during the 10-year period, and how it compares to not diversifying. Modern Portfolio Theory often had to rely on "static probability beliefs" and was limited by computational power (Markowitz, 1952). This study will build upon this by evaluating a much bigger dataset, a decade's worth of data, to help better capture a wide range of market behaviors. Furthermore, instead of assuming a normal distribution, this study moves beyond this to a simulation model using the multivariate t-distribution (MVT) to fully capture a more realistic simulation of the frequency of extreme events occurring further out in the tails. By doing this, the study is able to simulate how the diversification benefit works and behaves over an extensive period of time. By examining the underlying correlation structure, the study will also be able estimate and evaluate a theoretical systematic floor where individual asset risk is replaced by market risk. By identifying this floor, the study can evaluate how much of the unsystematic risk is eliminated with each added asset in the data, and determine a theoretical point in the data where the value of adding another asset becomes negligible.

## 2.3 Overview of planned analysis

The analysis will employ a Monte Carlo simulation using a multivariate t-distribution model to simulate asset returns using historical data from the chosen 10-year period. Using the standard deviation estimate derived from this model, the diversification effect can be determined by examining the rate of decrease in portfolio standard deviation with each added asset. Furthermore, the study will find a point in the data where the value of adding another asset becomes negligible, and the biggest part of the unsystematic risk from the model is diversified away. The diversification benefit of the found portfolio size will then be further studied empirically by a Monte Carlo simulation based on the historical data to see how it changed throughout the 10-year period.

## 3 Data

This section describes the dataset and provides an exploratory analysis to establish a foundation for the investigation.

### 3.1 Dataset

The dataset consists of daily stock returns for companies that were constituents of the S&P 500 index between the 13th of November 2015 and the 12th of November 2025. Data was retrieved via the R packages tidyquant (to obtain the list of S&P 500 constituents) and quantmod (to fetch the time series for each stock from Yahoo Finance). The S&P 500 index was selected to provide a broad and representative sample of the U.S. equity market.

From the initial list of 500 companies, stocks with incomplete data over the period were excluded. Missing data for a given stock was handled by removing that stock entirely from the sample, resulting in a final sample of 466 stocks. For each stock, there are 2,513 daily return observations, creating a balanced panel dataset with a total of  $466 \times 2,513 = 1,171,058$  observations. A transformation applied to the raw price data was the calculation of daily log returns, as defined by

$$r_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right)$$

where  $P_t$  is the closing price of a stock in time period  $t$ , and  $P_{t-1}$  is the closing price of a stock in time period  $t-1$ . This is a standard and necessary transformation in financial analysis. It converts non-stationary price series into stationary returns, enabling the additive aggregation of returns over time. The difference in log-returns approximates the daily percentage change in the stock price from one trading day to the next. For modeling, the data is organized into a matrix  $\mathbf{D} \in \mathbb{R}^{T \times I}$  where  $T = 2513$  is the number of trading days and  $I = 466$  is the number of stocks. Each element  $r_{t,i}$  of  $\mathbf{D}$  represents the daily log return of the stock  $i$  on day  $t$ . From this matrix, a variance-covariance matrix is estimated  $\mathbf{\Sigma} = Cov(\mathbf{D})$  where  $\mathbf{\Sigma} \in \mathbb{R}^{I \times I}$  (with dimensions  $466 \times 466$ ) is constructed.  $\mathbf{\Sigma}$  is defined as;

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho_{1,2}\sigma_1\sigma_2 & \dots & \rho_{1,466}\sigma_1\sigma_{466} \\ \rho_{2,1}\sigma_2\sigma_1 & \sigma_2^2 & \dots & \rho_{2,n}\sigma_2\sigma_{466} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{466,1}\sigma_{466}\sigma_1 & \rho_{466,2}\sigma_{466}\sigma_2 & \dots & \sigma_{466}^2 \end{bmatrix}$$

Where  $\rho_{i,j}\sigma_i\sigma_j$  is the covariance between two stocks ( $Cov(r_i, r_j)$ ) and  $\sigma_i^2$  is the variance of one stock. This matrix is fundamental to the study as it contains the individual variances of each stock along its main diagonal and the pairwise covariances in the off-diagonal elements. In total, the matrix captures  $\frac{466 \times (466-1)}{2} = 108,345$  unique pairwise covariances, providing a statistically robust foundation for simulating diversified portfolios.

### 3.2 Variables

- Independent Variable: Portfolio Size ( $n$ ): The independent variable is the number of equally weighted stocks in a simulated portfolio  $n \in \{1, 2, \dots, max_n\}$ . This variable was chosen as the experimental factor to isolate the effect of diversification. By systematically increasing  $n$ , the reduction of unsystematic risk can be precisely quantified and observed.
- Dependent Variable: Annualized volatility ( $\sigma_p$ ): The standard deviation of a portfolio's daily log returns. This variable was chosen to represent the risk of several equally weighted assets that belong to the same portfolio. and is defined as

$$\sigma_p = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_{p,t} - \bar{R}_p)^2} \times \sqrt{252} \quad (1)$$

Where the portfolio's daily return for day  $t$   $R_{p,t} = \mathbf{w}^T \mathbf{r}_t$  can be expressed as a scalar product where  $\mathbf{r}_t = [r_{1,t}, r_{2,t}, \dots, r_{n,t}]^T$  is the return vector for all  $n$  stocks for day  $t$ , and  $\mathbf{w} = [1/n, 1/n, \dots, 1/n]^T$  is the weight vector where equal weights are utilized. In finance, the annual volatility is often more commonly used than the daily; thus, for this report, the annual volatility will be used for investigation, thus a transformation of the volatility will be used. Since one year mostly includes 252 trading days, the volatility will be scaled by the square root of 252, i.e  $\sigma_{p_{annual}} = \sigma_{p_{daily}} \times \sqrt{252}$

The primary motivation for using these variables is to empirically determine the threshold at  $n$  at which the portfolio's total volatility converges toward the systematic risk. By observing the rate of decay in unsystematic risk, we can identify at which level of diversification the marginal benefit of adding more stocks becomes negligible.

### 3.3 Limitations

The dataset is considered sufficient for addressing the research question. It comprises daily return data for 466 individual stocks spanning ten years, covering multiple market regimes, including both calm and turbulent phases. The cross-sectional breadth ensures representation across a wide range of industries and firm characteristics, while the temporal dimension allows the analysis to average out short-run market-specific shocks. Together, this makes the dataset well-suited for studying how portfolio size affects total risk.

No additional firm-level variables are strictly required for the main objective of the study, since the focus is on the statistical relationship between the number of assets in a portfolio and aggregate risk, rather than on explaining returns or risk at the individual stock level. Nevertheless, the absence of explicit sector labels, firm size, or liquidity measures limits the possibility of conditioning on these characteristics in a more granular analysis.

Several potential lurking or confounding factors should be acknowledged. First, sector concentration may influence the observed relationship between portfolio size and risk. Portfolios with a small number

of assets may, by chance, be dominated by stocks from the same industry, leading to higher internal correlations and inflated portfolio risk that is not solely attributable to the portfolio size itself.

To mitigate this effect, portfolios are constructed using random sampling from the full universe of stocks, which helps average out sector-specific dependencies across repeated draws.

Second, survivorship bias may be present, as the dataset includes only stocks with complete return histories over the entire sample period. Firms that were delisted due to bankruptcy, mergers, or other events are excluded, potentially biasing the sample toward larger or more stable companies. This may result in an understatement of true portfolio risk, particularly for smaller portfolios, and should be considered when interpreting the results.

Finally, the choice of sample period itself may act as a confounding factor. Although the ten-year window includes varying market conditions, the results may still be influenced by period-specific features such as prolonged low-volatility environments or crisis episodes. As such, the findings should be interpreted as conditional on the selected time frame rather than as universally time-invariant properties of financial markets.

### 3.4 Exploratory analysis

This section presents an exploratory analysis of the S&P 500 returns data to establish empirical foundations for subsequent modeling. We examine three key aspects: (1) distributional properties, investigating if the normal approximation is appropriate; (2) correlation structures via network visualization to uncover clustering patterns and market segmentation; and (3) temporal dynamics of volatility and correlation using rolling windows to characterize market regimes and stress periods. These analyses provide critical insights that will be crucial for the rest of the study

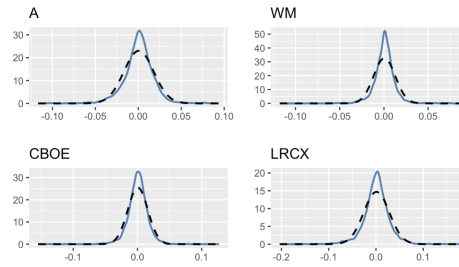


Figure 1: Univariate densities

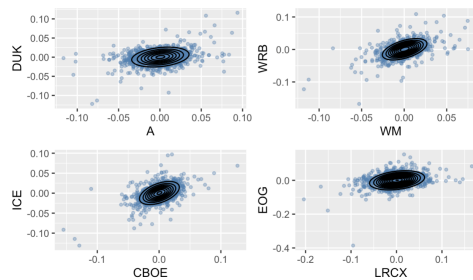


Figure 2: Bivariate Densities

In Figure 1, univariate densities are plotted for four randomly selected stocks from the S&P 500 index. The dashed line represents a theoretical normal distribution parameterized by the sample mean and variance. While the normal distribution provides a baseline approximation, the empirical data have a higher peak. This suggests that while daily returns often hover around the mean, extreme price movements occur more frequently than a Gaussian model would predict. Although the normal distribution appears to be a close match it underestimated the risk of large market shocks.

In Figure 2, the black ellipses represent a theoretical multivariate normal distribution, while the blue points represent the actual return data. In higher dimension, it seems the normal distribution matches the data worse than in the univariate case illustrated in Figure 1.

While the theoretical model predicts a neat elliptical shape, the actual returns exhibit fat tails, with many data points falling far outside the black rings. This shows that extreme movements, especially simultaneous price drops, occur much more frequently in the real market than the model suggests.

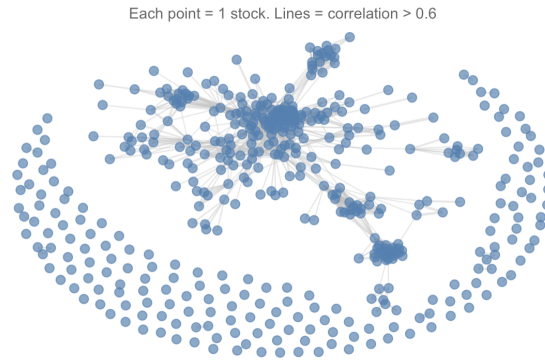


Figure 3: Correlation clusters

In Figure 3, the network visualization illustrates the correlation structure of the sampled S&P 500 stocks. Each node represents an individual equity, with edges indicating a correlation coefficient ( $\rho$ ) exceeding 0.6. A correlation cutoff of 0.6 was applied, representing approximately the 98th quantile of all pairwise correlations. A primary observation is the emergence of a dominant central cluster characterized by high node density and complex interconnectivity. This central component suggests a subset of stocks with significant co-movement. Surrounding this main body, several smaller, isolated subclusters are visible, alongside a substantial number of isolated nodes that exhibit no correlations above the specified threshold. Quantitatively, the topology reveals that approximately 40% of all stocks in the data do not belong to the primary central cluster. Overall, this visualization highlights a clear division in the market's structure: while a large majority of stocks move in sync within a central cluster, a significant portion remains decoupled, suggesting that not all equities are driven by the same market forces.

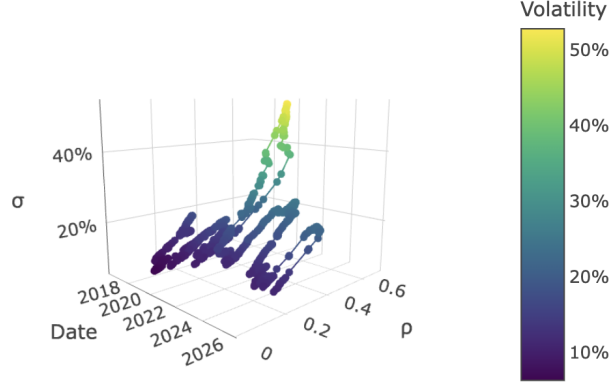


Figure 4: 3D Timeline of correlation and St.dev of S&P 500

In Figure 4 , one can observe the movements of the mean pairwise correlation ( $\rho$ ) and the mean volatility ( $\sigma$ ) for the dataset for the studied period. The means are calculated using a rolling window of 120 days, chosen to provide a balance between capturing meaningful medium-term trends and filtering out short-term market noise. Both  $\rho$  and  $\sigma$  are dynamic over the entire period, exhibiting a visible relationship. The most spectacular feature in the plot is the peak in 2020, representing the COVID-19 crash, where  $\sigma$  and  $\rho$  reached their highest values between the period 2015-2025, at 53% and 0.61, respectively. The time series moves in waves, where each peak represents periods of market instability and heightened uncertainty, such as the inflationary crisis in 2022 and the geopolitical tensions in 2025.

## 4 Method

This section outlines the methodology that was used throughout the study, including the main MVT model simulation and the empirical simulation evaluating the diversification benefit during the 10-year period.

### 4.1 Investigated relationships

The main relationship that was studied in the report is the relationship between the number of assets in the portfolio ( $n$ ) and the annualized portfolio volatility ( $\sigma_p$ ). More specifically, the decay rate of the portfolio was studied as the number of simulated assets grew, along with the threshold where the value of adding another asset became negligible. Furthermore, the study also investigated the role of asset correlation and how this affected the relationship between the number of assets in the portfolio ( $n$ ) and the portfolio volatility ( $\sigma_p$ ) during different market regimes.

### 4.2 Multivariate t-distribution

While the multivariate normal distribution offers tractability for portfolio modeling, empirical evidence strongly suggests that asset returns exhibit non-normality, particularly in the form of heavy tails. The multivariate Student-t distribution is particularly well-suited for modeling the fat-tailed behavior observed in financial return data (Harvey et al., 2010)

The MVT model is constructed as:

$$R \sim MVT(\boldsymbol{\mu}, \boldsymbol{\Sigma}^*, \nu)$$

where

$\boldsymbol{\mu}$  represents the location vector, which is the central tendency of the distribution. Since the focus of this study is to analyze the diversification through the standard deviation of the portfolio, this estimate is not of interest in this analysis, and therefore, it is set to 0. This ensures that the reduction seen in the standard deviation is purely the result of the co-movement between the assets and not distorted by an estimation of this parameter.

$\boldsymbol{\Sigma}^*$  is the scale matrix and  $\boldsymbol{\Sigma}$  is the covariance matrix. Since the variance of the t-distribution gets inflated by fat tails, it is necessary for the MVT package that the input parameter is the scaled matrix to match the historical S&P data. To adjust for this inflation, the input is the scaled covariance matrix, with the following formula used to derive the scaled covariance matrix:

$$\boldsymbol{\Sigma}^* = \boldsymbol{\Sigma} \frac{\nu}{\nu - 2}$$

$\nu$  is the parameter estimated by MLE through the method EMCE, which will be explained further. This is done to ensure that it matches the frequency of extreme events occurring in the data and determines the thickness of the tails in the distribution.

### 4.3 ECME

To ensure that the degrees of freedom ( $\nu$ ) that are used for the MVT model reflect the actual market conditions through the frequency of extreme events, the degrees of freedom have to be empirically determined, since they cannot be easily observed.

Since the model uses a single degree of freedom parameter for the model, one value was needed for the entirety of the portfolio. In this study, the degrees of freedom estimator was estimated using an iterative maximum likelihood method through an Expectation-Maximization (EM), more precisely, an expanded version, an ECME algorithm, established by Liu & Rubin (Liu and Rubin, 1995). This was done using the `fitHeavyTail` package in R (Palomar et al., 2023). This method is motivated since it provides robustness by handling outliers, which occur frequently in financial data. It is also efficient in estimating the degrees of freedom parameter. The algorithm works by identifying a robust scatter matrix and the degrees of freedom; we will only extract the degrees of freedom from this. First, in the expectation (E) step, the algorithm assigns each historical observation an expected weight based on its Mahalanobis distance from the center of the multivariate t-distribution. Extreme outliers are given lower weights. Then, in the CM-step 1, the scatter matrix and center are updated using weighted least squares based on these weights. Finally, in the CM-step 2 step, by maximizing the likelihood of the observations directly, the degrees of freedom are updated based on the scatter matrix (Liu and Rubin, 1995). When this is done, the study extracted the final estimated degrees of freedom. This is more effective and faster than the regular EM algorithm, which would update the degrees of freedom parameter based on the expected weights, requiring the algorithm to look at the weight step, forcing the parameters to change more gradually. Using the ECME therefore requires fewer computations by being able to look at the observations directly when updating the degrees of freedom parameter (Liu and Rubin, 1995).

## 4.4 Monte Carlo simulation

With the parameters for the model fully established, a Monte Carlo simulation was established with the calibrated MVT used as the generative engine. A Monte Carlo model is a method that uses repeated sampling to estimate the outcomes of an event. By using a set of input values, the model creates a distribution and then recalculates the results repeatedly. Monte Carlo relies on the law of large numbers, where repeated random sampling will make the simulated average converge towards the theoretical sample (Glasserman, 2003).

The main variable calculated and analyzed through the MVT model is the annualized portfolio volatility ( $\sigma_p$ ). This variable is derived from the mean daily return variable, which represents the daily return of an equally weighted portfolio. For each asset size (1:100), the loop will pick the scaled adjusted covariance matrix, but cut out a matrix only containing the chosen stocks and their specific scaled covariances. The MVT model then generates simulated daily returns for 252 trading days (1 year) for each chosen stock based on the scaled matrix, 1000 times. The mean asset returns for each day are then calculated, which gives us the mean return of an equally weighted portfolio. This is then used to calculate the annualized volatility. Since we perform 1000 iterations, we take the mean of this, giving us the mean annualized volatility for each portfolio size. The annualized volatility is the dependent variable, which is used to measure the diversification effect. By evaluating this variable through different portfolio sizes, it can be evaluated how diversification works and what happens to the volatility as the asset grows and the market conditions change. The purpose of the model was to study the diversification rate as the portfolio size increased, which will be further explained later in the analysis. Furthermore, a threshold where 95% of the unsystematic risk had been eliminated was then calculated.

After the threshold size was found, a second Monte Carlo simulation was implemented to evaluate how a randomly sampled number of assets using this size found in the MVT model would perform compared to one asset, from the data during the chosen time period. Instead of drawing returns from the MVT model, this simulation drew data directly from the historical data, using 500 iterations. This asset size was then evaluated for the 10-year period by studying how the annualized portfolio volatility ( $\sigma_p$ ) behaved along with the correlation of the assets during the different market regimes.

While both the Monte Carlo simulation and a historical simulation would be based on historical data, a historical simulation would be constrained by the events that occurred during the chosen period. A Monte Carlo simulation is preferred because it allows for robustness by simulating thousands of scenarios based on the structure of the data, rather than just one single event. By doing this, the study ensures that the estimates are statistically robust. By using a large number of iterations, the small noise observed in the time period will also be able to be canceled out, in favor of the true underlying diversification effect and risk structure to be studied.

## 4.5 Definition of Portfolio and Risk

A portfolio is defined for this report as a combination of stocks, and for this study, all the stocks that are included in a specific portfolio are equally weighted, i.e, each stock is  $\frac{1}{n}$  of the proportion of the whole portfolio. To estimate the risk of a portfolio, volatility is often used, which is defined as the standard deviation of the returns from this portfolio. This risk can be separated into two parts: systematic risk and unsystematic risk  $\sigma_{total} = \sigma_{systematic} + \sigma_{unsystematic}$ . The unsystematic risk is the risk that is unique to each firm. This is the risk that can be eliminated by adding more assets; therefore, it is also often referred to as the diversifiable risk. The Systematic Risk ( $\sigma_{syst}$ ), often referred to as market risk, is the non-diversifiable component of the total portfolio risk (Bodie, Kane, and Marcus, 2013).

Recall from Equation 1 that portfolio volatility is based on the variance of portfolio returns. Expressing the portfolio return as  $R_{p,t} = \mathbf{w}^\top \mathbf{r}_t$  implies that the portfolio variance can be written as  $\mathbf{w}^\top \Sigma \mathbf{w}$ . Under equal weighting, this expression naturally decomposes into variance and covariance terms, corresponding to unsystematic and systematic risk, respectively. Which can be expressed as

$$\sigma_p = \sqrt{\frac{1}{n} \sum_{i=1}^n \frac{1}{n} \sigma_i^2 + \sum_{j=1}^n \sum_{\substack{i=1 \\ j \neq i}}^n \frac{1}{n^2} \text{Cov}(r_i, r_j)} \quad (2)$$

If we define the average volatility and the average covariance of the stocks as  $\bar{\sigma}^2 = \frac{1}{n} \sigma_i^2$  and  $\overline{Cov} = \frac{1}{n(n-1)} \text{Cov}(r_i, r_j)$  we can rewrite the equation as

$$\sigma_p = \sqrt{\frac{1}{n} \bar{\sigma}^2 + \frac{n-1}{n} \overline{Cov}} \quad (3)$$

As illustrated by the decomposition, risk that belongs to each specific firm (unsystematic risk) vanishes as  $n$  increases, while the second represents systematic risk that remains as  $n$  increases. Consequently, total portfolio volatility depends increasingly on the average covariance between assets. In this study, the systematic risk ( $\sigma_{\text{syst}}$ ) serves as the 'risk floor' for the analysis, where  $\sqrt{\overline{Cov}} \approx 18.6\%$ , we define this value as the overall market risk for this study. The floor represents the asymptotic limit as the number of assets approaches infinity, where all unsystematic risk has been eliminated. This point cannot be fully reached unless all assets in the entire market are owned.

## 4.6 Convergence threshold

The optimal amount of assets will be calculated by determining the point where 95% of the unsystematic risk has been eliminated.

The threshold is calculated as follows:

$$\sigma_{\text{threshold}} = \sigma_{\text{systematic}} + (\sigma_{n=1} - \sigma_{\text{systematic}}) \times 0.05$$

where

$\sigma_{n=1}$  represents the average annualized volatility for a single asset. This therefore represents the total risk, both unsystematic and systematic, before any diversification has occurred. This represented an annualized volatility of around 31.8%.

By taking the difference between  $\sigma_1$  and the systematic risk floor, we are able to calculate the unsystematic risk, the risk that can be diversified away by adding more assets. This represented around 13.1% of the annualized volatility. Finally, the threshold is found at the point where only 5% of the unsystematic risk remains. The threshold was found at 19.2%. The threshold is then identified as the smallest  $n$  for which the simulated portfolio volatility is less than or equal to  $\sigma_{\text{threshold}}$ .

The threshold was chosen since a portfolio's volatility can theoretically never be equal to the systematic floor, since this would require owning all available assets in the market. Therefore, a point where all unsystematic risk is eliminated does not exist in a real market for finite portfolios.

However, by choosing the threshold of 95%, a precise number of assets from the data where a majority of the unsystematic risk is gone can be found, where the value of adding another asset becomes negligible. We assume that for a typical investor, the benefit of the last few percentages and the minor benefit of these support the threshold to be cut off at 95. This measure does not serve as a universal constant but rather a threshold that can be evaluated and applied for the specific time period and dataset. Furthermore, the convergence point is subject to both noise from the historical data and random sampling variation from the chosen simulation seed. However, finding a convergence point for the chosen data serves as an appropriate approximation for studying risk behavior and allows for a comparison between this number and a single asset during the chosen time period, to further compare and evaluate diversification benefits during the whole time period and different market regimes.

## 4.7 Model evaluation

The model evaluation was performed through two steps. First, the presence in the data of extreme events, "fat tails", was evaluated by performing the chosen ECME approach from Chapter 4.3. After this, if the estimate indicated signs of fat tails, a QQ-plot was then performed to evaluate if the data provided a good fit to the multivariate t distribution. The chosen ECME method provides a reliable estimate of degrees of freedom. The estimate of the degrees of freedom also gets robust against outliers by giving outliers a lower weight.

The ECME approach was first performed on the data, which resulted in the observed value of around 5.968, showing empirical evidence of fat tails. This indicated that the multivariate t-distribution would be a better choice than a multivariate normal distribution, which would underestimate the presence of the extreme events in the data. The ECME model also has stable monotone convergence, meaning the model's fit to the data will improve with each iteration. This ensures that a reliable degree of freedom estimate will be reached. (Liu and Rubin, 1995). However, to further evaluate and ensure that the data fit a multivariate t distribution with the estimated degrees of freedom parameter, QQ-plots for different asset sizes were plotted.

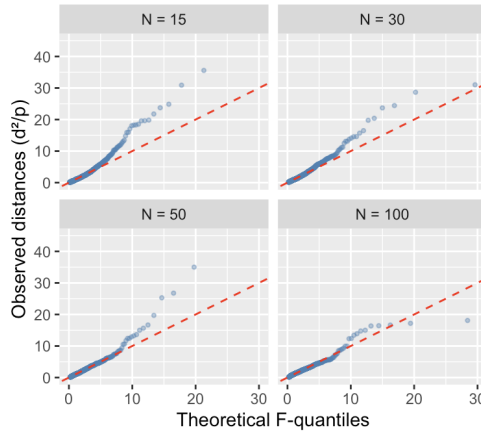


Figure 5: QQ plots of chosen asset sizes

Figure 5 presents a comparative diagnostic analysis of the data's distribution using Mahalanobis distance QQ-plots for the Multivariate Student's t-distribution (MVT) for asset sizes 15, 30, 50, and 100.

Since we cannot plot all dimensions at once, the Mahalanobis distance allows us to summarize the joint behavior into a single dimension. For each portfolio size  $n$ , we calculate the squared Mahalanobis distances  $d^2$  relative to the location vector (set to 0 in our model), which is then weighted by the scale matrix used in our model. The plot compares the observed distances against an F-distribution. This was chosen since it represents the theoretical distribution of the scaled Mahalanobis distances.

Looking at the QQ-plots for the chosen asset sizes, it can be seen that the model provides a good fit for most of the data. While the data does not perfectly align with the MVT model, exhibiting some residual dispersion in the far upper tail for the most extreme events, mostly seen in asset size 15, the fit becomes increasingly more stable as the asset size increases. As in most cases, even if the chosen model is not a perfect fit for all points, the model and the degrees of freedom are still deemed as a reasonable fit for capturing the essential risk characteristics for the analysis.

Another input to the MVT-model besides just the degrees of freedom is also the scale matrix ( $\Sigma^*$ ) that depends on the covariance matrix ( $\Sigma$ ). The Cross-sectional dependence is captured entirely through the empirically estimated from S&P 500 returns over the period 2015–2025. This implicitly assumes that the average dependence structure observed over the sample period is representative of a typical market environment. While co-movements and variances are known to vary over time, which can also be seen in Figure 4, the model is not intended to describe short-run dynamics but rather to provide a stylized representation of long-run diversification properties.

The model assumes homogeneity in tail behavior by imposing a single degrees-of-freedom parameter across all assets. However, as illustrated in Figure 5, the model does not perfectly capture the empirical tails, meaning the probability of extreme events is not perfectly symmetrical. This simplification is justified by the portfolio-level focus of the analysis, as diversification predominantly depends on the aggregate correlation structure rather than the idiosyncratic tail behavior of individual assets. Consequently, the model’s performance is evaluated based on its ability to capture portfolio-level risk dynamics.

In summary, while the MVT model does not capture every nuance of the S&P 500’s complexity, it accounts for the heavy-tailed nature of financial returns which a model like the MVN would not be able to capture. The observed deviations in the extreme tails suggest that results should be interpreted with an awareness of the model’s symmetric constraints. However, for the purpose of analyzing how portfolio volatility scales with size, the model remains a highly appropriate tool. The simplification of tail symmetry is a necessary trade-off that allows for a stable and consistent estimation of diversification effects across varying portfolio dimensions, without introducing the over-parameterization associated with more complex models.

## 5 Implementation

### 5.1 Package and main function

The core statistical tool used in this study is the R-package `mvtnorm` version 1.3-3 (Genz and Bretz, 2009). This provides functionality for density evaluation, distribution functions, and random number generation from multivariate normal and multivariate t-distributions. The package is particularly useful where joint dependence structures and heavy-tailed behavior are essential features of the data, which tend to occur more often in the financial market (Mandelbrot, 1967).

The main advantage of `mvtnorm` is its ability to simulate observations from a multivariate t-distribution, denoted  $MVT(\boldsymbol{\mu}, \boldsymbol{\Sigma}^*, \nu)$ , where  $\boldsymbol{\mu}$  is the location vector,  $\boldsymbol{\Sigma}^*$  is the scale matrix, and  $\nu$  is the degrees of freedom parameter controlling tail thickness. Compared to a multivariate normal model, the multivariate t-distribution offers substantially greater flexibility by allowing extreme joint movements to occur with higher probability. This makes `mvtnorm` particularly appropriate for Monte Carlo simulations aimed at evaluating portfolio risk and diversification effects.

The primary function used from the package is `rmvt()`, which generates random samples from a multivariate Student's t-distribution. The function offers a flexible interface with several arguments that allow explicit control over distributional shape, dependence structure, and numerical stability. In this study, only a subset of these arguments is used directly, while others are deliberately left at their default values to match the modeling objectives.

The core function call is

```
rmvt(t, sigma, df, delta, type)
```

where the argument `t`, which specifies the number of observations generated per simulation is set to 252 to represent one trading year. This choice ensures that simulated portfolio risk is directly comparable to empirically observed annualized risk measures. (While the function `rmvt()` uses `n` to denote the number of observations, this report will refer to this parameter as `t` to avoid confusion with the number of stocks in a portfolio, which is defined as `n`).

The argument `sigma` defines the scale matrix  $\boldsymbol{\Sigma}^*$  and governs the entire cross-asset dependence structure of the simulated returns.  $\boldsymbol{\Sigma}^*$  is estimated from the empirical data, allowing the simulation to reflect realistic correlation patterns between assets. This choice is central to the study, as the effect of portfolio size on total risk is driven primarily by the interaction between cross-sectional dependence and diversification.

The degrees of freedom parameter `df` controls the heaviness of the tails of the distribution, which in this study is set to approximately 5.96, fetched from MCME. By selecting finite values of `df`, the simulation departs from the Gaussian benchmark and allows for excess kurtosis and an increased probability of joint extreme outcomes.

Several additional arguments are available, but are intentionally fixed at their default values. The non-centrality parameter `delta` is set to its default value zero, corresponding to a central multivariate t-distribution. This reflects the focus on modeling return variability and dependence rather than systematic mean shifts. Similarly, the argument `type` is left at its default value, since in the central case the alternative parameterizations coincide.

Overall, the flexibility of `rmvt()` lies in its modular structure: dependence (via `sigma`), tail behavior (via `df`), and time aggregation (via `n`) can be varied independently. This makes the function particularly well-suited for Monte Carlo experiments aimed at isolating the impact of modeling assumptions on portfolio risk and diversification.

## 5.2 Simulation Design for Package Validation

The simulation study is designed to evaluate whether the function `rmvt` from the package `mvtnorm` produces outputs consistent with theoretical expectations under known dependence structures. Data are generated from an idealized underlying model with known theoretical properties. We construct controlled data-generating processes for which the expected behavior is known analytically.

Specifically, we adopt a parametric Monte Carlo design where synthetic returns are generated from multivariate t-distributions under two extreme and theoretically transparent dependence structures. The output from the simulation from these settings will then be evaluated against theoretically known results to investigate if `rmvt` gives reasonable output.

To construct these ideal models and ensure an exact validation against the theoretical benchmarks, we transform the empirically estimated covariance matrix to be entirely homogeneous. This is achieved by replacing each individual asset's variance with the market's average variance ( $\sigma^2 = \bar{\sigma}^2$ ), effectively eliminating statistical noise, where the correlation ( $\rho$ ) will be manually adjusted to observe two different extreme events.

1. **Perfect correlation matrix  $\Sigma_{\rho=1}$ :** All assets are assumed to move in perfect unison. The off-diagonal elements are therefore set equal to, implying a pairwise correlation of exactly one:

$$\Sigma_{\rho=1} = \begin{bmatrix} \sigma^2 & \sigma^2 & \dots & \sigma^2 \\ \sigma^2 & \sigma^2 & \dots & \sigma^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma^2 & \sigma^2 & \dots & \sigma^2 \end{bmatrix}.$$

2. **Independence matrix  $\Sigma_{\rho=0}$ :** All assets are assumed to be mutually independent. The off-diagonal elements are therefore set to zero:

$$\Sigma_{\rho=0} = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}.$$

By generating data from these ideal models and comparing the simulation output to the corresponding theoretical predictions, we can assess whether the `mvtnorm` package correctly reproduces the intended correlation structure and yields results consistent with portfolio theory.

### 5.2.1 Simulation Procedure

For each covariance structure ( $\Sigma_{\rho=1}$  and  $\Sigma_{\rho=0}$ ) and for each portfolio size  $n = 1, 2, \dots, 100$ , the following steps are repeated 1 000 times:

1. Define a  $n \times n$  scale matrix  $\Sigma_{\rho}^*$  for the chosen dependence structure for the current portfolio size  $n$ .
2. Simulate daily returns for 252 trading days using

$$\text{rmvt}(\mathbf{t} = 252, \text{sigma} = \Sigma_{\rho}^* \in \{\Sigma_{\rho=0}^*, \Sigma_{\rho=1}^*\}, \text{df} = \hat{\nu}),$$

3. Construct an equally weighted portfolio return series.
4. Compute the annualized portfolio volatility using Equation 1

The mean volatility over the 1 000 replications is recorded for each  $n$ .

### 5.2.2 Theoretical Benchmarks

Before analyzing the simulation output, theoretical results that should be expected under the idealized dependence structures considered above are presented. These analytical benchmarks provide a reference against which the Monte Carlo results from `rmvt` can be evaluated.

Recall Equation 2, under the simplifying assumption of homogeneous asset variances ( $\sigma_i^2 = \sigma^2$  for all  $i$ ) and a constant pairwise correlation  $\rho$  between all assets, the volatility of an equally weighted portfolio can be expressed as

$$\sigma_p = \sqrt{\frac{1}{n}\sigma^2 + \frac{n-1}{n}(\rho \cdot \sigma^2)}. \quad (4)$$

Equation (4) provides a transparent decomposition of portfolio risk. While the first term vanishes as the number of assets increases, the second term remains whenever assets are correlated.

From (4), two extreme cases follow directly and serve as theoretical validation benchmarks for the simulation study:

- **Perfect correlation** ( $\rho = 1$ ):

$$\sigma_p = \sqrt{\frac{1}{n}\sigma^2 + \frac{n-1}{n}\sigma^2} = \sqrt{\sigma^2} = \sigma. \quad (5)$$

In this case, portfolio volatility is constant and independent of the number of assets  $n$ . Since all assets move in perfect unison, diversification provides no risk reduction.

- **Zero correlation** ( $\rho = 0$ ):

$$\sigma_p = \frac{\sigma}{\sqrt{n}}, \quad (6)$$

implying that portfolio volatility decays at a rate  $1/\sqrt{n}$ . This represents the maximal diversification benefit achievable under independence.

These analytical results define the expected behavior of portfolio volatility under the two idealized covariance structures used in the simulation. Consequently, if the `mvtnorm` package correctly implements the multivariate t-distribution with the specified dependence structure, the Monte Carlo results should satisfy the following:

- Under  $\Sigma_{\rho=1}$ , portfolio volatility should remain approximately constant across all portfolio sizes  $n$ .
- Under  $\Sigma_{\rho=0}$ , portfolio volatility should decrease proportionally to  $1/\sqrt{n}$ .

Observed deviations from these patterns would indicate that the simulated data do not reflect the intended correlation structure, thereby calling into question the reliability of the package for applied risk analysis.

## 5.3 Simulation results

This section presents the results from the simulation study based on the idealized dependence structures. By comparing the Monte Carlo outputs against the analytical benchmarks, we evaluate whether the simulated data align with theoretical expectations, thereby validating the reliability of the `mvtnorm` package for financial risk modeling.

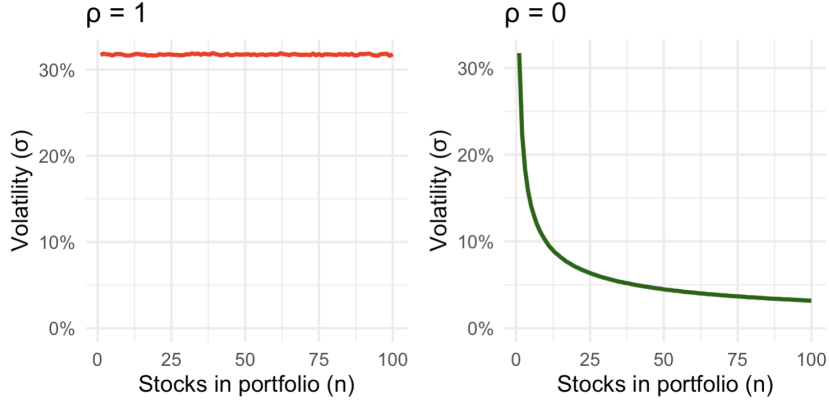


Figure 6: Relationship between number of assets and mean sample volatility, ( $\rho = 1$  vs  $\rho = 0$ )

As can be seen in Figure 6, the standard deviation seems to be at a constant rate when  $\rho = 1$ . In other words, the risk does not get reduced when  $n$  increases, which is in line with the expectations.

In the other case, when all stocks are independent, we observe a pretty smooth line that approaches zero at what seems to be the rate of  $1/\sqrt{n}$  which  $n$  increases. To further evaluate if the simulation gives reasonable results, numerical results will also be presented.

| n   | Simulated $\sigma_{\rho=1}$ | Theoretical $\sigma_{\rho=1}$ | Simulated $\sigma_{\rho=0}$ | Theoretical $\sigma_{\rho=0}$ |
|-----|-----------------------------|-------------------------------|-----------------------------|-------------------------------|
| 1   | 31.8%                       | 31.8%                         | 31.8%                       | 31.8%                         |
| 10  | 31.7%                       | 31.8%                         | 10%                         | 10.1%                         |
| 50  | 31.8%                       | 31.8%                         | 4.5%                        | 4.5%                          |
| 100 | 31.7%                       | 31.8%                         | 3.2%                        | 3.2%                          |

Table 1: Simulation results

As can be observed in Table 1, a comparison between the simulated results and the theoretical results is presented. The theoretical results are computed via Equation 5 & 6. The results are reported for increasing portfolio sizes  $n$ . The close agreement between simulated and theoretical values indicates that the Monte Carlo procedure accurately reproduces the theoretical diversification effects implied by the assumed correlation structures.

Overall, the implementation using the package `mvtnorm` and the function `rmvt()` provides a flexible and theoretically sound framework for studying diversification. The ability to explicitly control tail behavior and correlation structure makes the package particularly suitable for investigating risk convergence and robustness in financial portfolios.

## 6 Results

### 6.1 Variability of the standard deviation estimate

Since the annualized volatility is used extensively to study the effect of diversification, it is important to evaluate the uncertainty in this estimate.

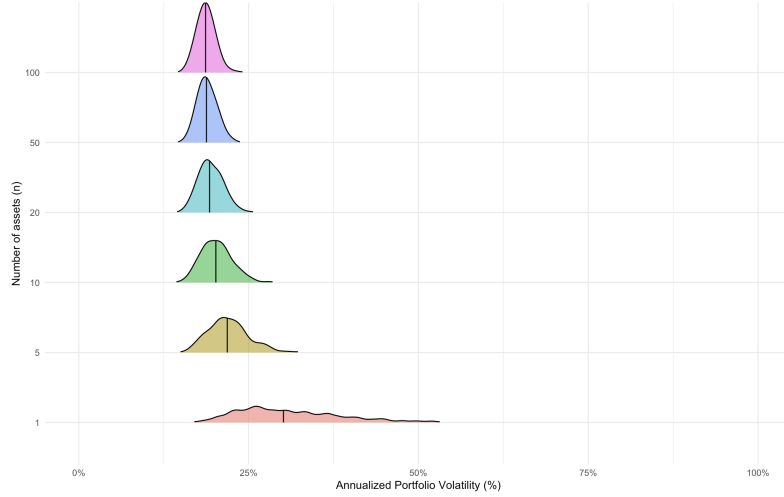


Figure 7: Variability of portfolio standard deviation estimate

In Figure 7 the variability for each annualized volatility at different asset sizes can be seen. At the lowest level,  $n = 1$ , the distribution is wide and flat. This illustrates the variability of volatility across the available stocks from the S&P 500. Any stock from the market could be chosen here, which causes the volatility to range from around 15% to around 50% since both high and low volatility stocks will be captured by the simulation. The estimate quickly stabilizes as  $n$  increases since each trial will become an average of many different stocks. At  $n = 5$ , the variability of the estimate has already stabilized dramatically. The variability then becomes even smaller with each  $n$ . By 20 stocks, the distributions already look very similar to each other as  $n$  grows. This indicates that an increasing number of  $n$  will also reduce the uncertainty of the risk estimate.

While the number of repetitions in the simulation ensures the smoothness and reduction of noise, the reduction in the spread with each increased  $n$  is a result of the individual asset risk taking out each other, resulting in a smoother and predictable curve when more assets are added. Increasing the number of iterations in the simulation would not affect the risk dispersion between 1 and 100 assets, but rather reduce the noise even further and provide an even more stable result.

## 6.2 Convergence of the annualized volatility towards the systematic floor

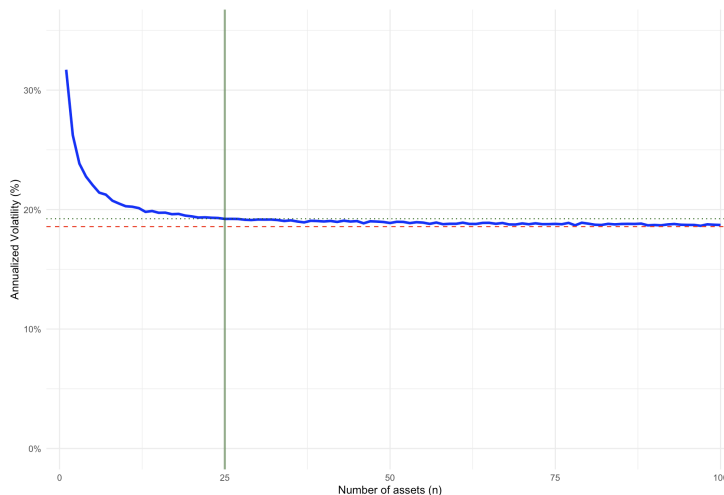


Figure 8: Relationship between the number of assets and annualized portfolio volatility

Figure 8 illustrates the results from the MVT model. The figure illustrates the number of assets along with their annualized volatility as the blue line, together with the found convergence point on the green line at  $n = 25$ . The red line illustrates the systematic risk floor, which is the risk that cannot be eliminated. The figure shows a clear non-linear decay in the annualized volatility towards the red line.

The plot clearly visualizes the effect of diversification. At the beginning of the plot, from 1 asset, there is a sharp decline in mean sample standard deviation as the number of assets increases, indicating the effect of the individual asset risk of the stocks canceling each other out. As  $n$  grows, it can be observed how the decline of the annualized volatility slows down while slowly approaching the systematic risk, where the line stabilizes. This illustrates that eventually, after adding enough assets, almost all of the unsystematic risk has been eliminated, and the volatility has approached the systematic floor, where the volatility continues to lie around the same percentage, even with more assets added.

After performing the convergence test, the number of assets where 95% of the unsystematic risk is eliminated was found at  $n = 25$ .

Table 2: Achieved diversification per number of assets

| Number of assets | Annualized Volatility % | Achieved diversification (%) |
|------------------|-------------------------|------------------------------|
| 1                | 31.8                    | 0                            |
| 5                | 22                      | 74                           |
| 25               | 19,2                    | 95                           |
| 50               | 18,9                    | 98                           |
| 100              | 18,7                    | 99                           |

Table 2 shows how much of the systematic risk was eliminated for different numbers of  $n$ . The achieved diversification metric calculates the percentage of unsystematic risk eliminated relative to the theoretical systematic floor. At  $n = 1$ , the volatility is around 31.8%. By adding just 4 assets, it can be seen how the achieved diversification reaches around 74%, with the annualized volatility dropping to around 22%. At the found convergence point of 25 assets, around 95% of the systematic risk is eliminated, with the portfolio volatility dropping to around 19.2%.

Beyond this point, it can be seen how the marginal benefit of adding more assets diminishes drastically. By 50 assets, the volatility has only dropped by around 0.3 percentage points, and the eliminated unsystematic risk has gone from 95 to 98 percent. By 100 assets, the annualized volatility has dropped by 0.2 percentage points from 50 assets; here, approximately 99% of the unsystematic risk has been eliminated. This illustrates that each new asset has a dramatically lower effect after our threshold, similar to what was seen in Figure 8. In the beginning, each new asset will have a more powerful impact by helping canceling out the volatile movements, but eventually, after enough stocks have been added, this effect will slowly level off.

As seen in (3), earlier, since the systematic floor is a theoretical benchmark that represents a point where  $n$  has reached infinity and all unsystematic risk is gone, we will never fully reach 100% eliminated unsystematic risk since we test a finite set of assets.

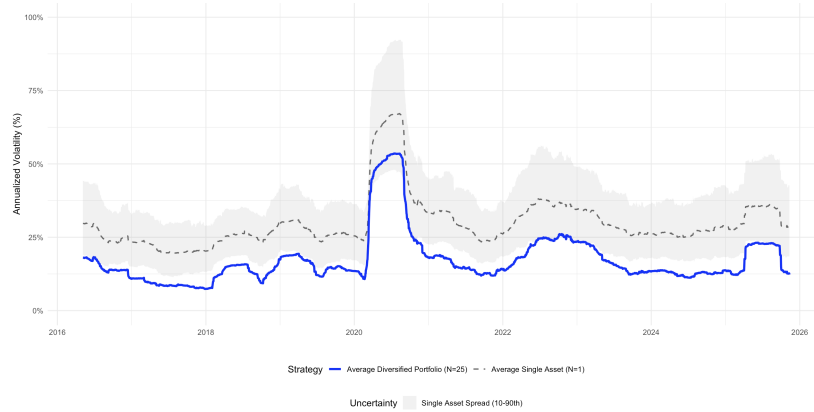


Figure 9: Time series of annualized volatility: 1 single asset vs 25 assets

To evaluate the found threshold and capture the dynamics of volatility without having too much noise in the plot, a 120-day rolling window was applied to calculate the average annualized volatility and the average pairwise correlation for a 25-stock portfolio vs a single stock portfolio from the original dataset, for the 10-year period.

This was then compared to a portfolio with a single asset. To ensure robustness and avoid sampling bias, the 25 stocks were sampled 500 times, and one stock was then drawn from this for a valid and representative comparison to the same "universe" of assets, rather than again sampling one stock.

The time series in Figure 9 illustrates the average volatility of 25 randomly sampled stocks compared to a single sampled stock. To illustrate the possible risk and variability taken with only one stock, the 10th and 90th percentiles for the single stock across the 500 simulations were also plotted. The figure illustrates that the chosen portfolio size of 25 assets almost consistently displays an advantage over a single stock with a lower average volatility. The figure displays the advantage that diversifying has, most notably during the normal market periods. However, even during the extremely volatile periods like the COVID-19 crash, the diversified portfolio shows a lower average volatility compared to the single stock in most cases, and dampens the volatility. The time series also shows the extreme volatility a single stock can reach, especially during the more volatile market periods.

Even if the average volatility of 25 stocks almost consistently manages to achieve a lower volatility than the single asset, the time series shows that the larger portfolio does not lead to immunity to market forces. While 25 stocks eliminate most of the noise of the unsystematic risk during normal market periods, the diversified portfolio will still be affected by the heavy market movements where the market-wide risk becomes the dominant force, rather than the unsystematic risk. During these periods, the diversification fails to provide protection and will still illustrate an increasingly higher volatility. This shows that even if diversification is effective in normal market conditions, it does not offer the same protection in periods of financial instability with heavy market movements. The effect of individual assets canceling the movements of each other starts to fail when the assets instead move in unison and when the average pairwise correlation rises substantially, which was also seen in Figure 4 in the data section.

| Regime                                                 | Days | Mean Vol Ratio (%) |
|--------------------------------------------------------|------|--------------------|
| Low correlation (Stable) $\rho < 0.20$                 | 494  | 43.5               |
| Medium Correlation (Neutral) $0.2 \leq \rho \leq 0.35$ | 1272 | 53.7               |
| High Correlation (Crisis) $\rho > 0.35$                | 628  | 67.8               |

This relationship can be further shown in Table 3. Like what was seen in the data section in Figure 4, during volatile market periods, both the annualized volatility and the pairwise correlation tend to go up. By illustrating the connection between volatility and correlation, the diversification effect can be further studied during different market regimes.

Using the same simulation results, the rolling average volatility ratio was calculated between the 25 randomly sampled stocks and a single stock. By doing this, the relationship between the rolling correlation and the rolling average annualized volatility could be further illustrated. The market was divided into three market regimes based on the 25th and 75th quantiles of the correlation levels between the stocks for the market days.

In the low regime, which represented around 494 trading days, the volatility ratio was the lowest, around 43.5, meaning the portfolio retained around 44% of the risk of an average stock. The diversification, therefore, had the best effect during the low correlation regime, which goes in line with the main results theory of diversification, that adding more assets less correlated to each other will diminish the overall portfolio volatility.

Moving to the regime with the medium correlation between 0.2 and 0.35 resulted in a higher volatility ratio. The portfolio now increased to retaining around 53.7% of the risk of an average stock, showing that the diversification effect had gotten increasingly worse.

Finally, in the high correlation regime where the pairwise correlation was the highest in the data, the volatility ratio was the highest. At this point, the portfolio retained around 68% of the risk of an average stock, which shows that the diversification benefit had drastically decreased. Just like what was shown in the graph during the extreme events, the effectiveness of the diversification dramatically diminished during the heavy market movements, when diversification starts to fail, and when assets, instead of canceling each other out, start to move in unison.

## 7 Conclusion

The goal of the study has been to study the rate of systematic risk convergence through the MVT model, and further, to study the diversification benefit for the chosen 10-year market period. The results illustrate that diversification works as a helpful tool for investors to achieve lower portfolio volatility and a more stable portfolio.

By using the MVT model through the Monte Carlo simulation, the study managed to identify and validate a robust model for estimating the rate of risk reduction from the S&P 500 from 2015 until 2025, where an increased amount of assets showed a rapid decay in annualized volatility towards the systematic risk, proving the main result of diversification. By using a controlled data-generating process with predefined correlation parameters, the study could, with financial theory, confirm that the chosen package worked as expected. The observed ridge plot illustrated that by adding more  $n$ , the extreme outliers in the yearly standard deviation were eliminated. The diversified portfolio with a higher amount of assets allowed for a more predictable standard deviation, regardless of the chosen stocks. The findings of the analysis confirm that, based on the MVT model, a convergence point where 95% of the unsystematic risk could be eliminated was found at  $n = 25$ . The results showed that adding stocks beyond this point was negligible, with the volatility only decreasing by around a half percentage point between 25 and 100 assets in the portfolio.

The threshold could then be evaluated empirically by using the original dataset for the 10-year period. By evaluating the rolling volatility and pairwise correlation for the found threshold at 25 and a single simulated asset, the analysis demonstrated that the diversification benefit was driven by two primary factors: the number of assets and their underlying correlation. The time series illustrated that 25 assets consistently showed a more stable result across the 10-year time period, while the variability in having a single asset was high. However, an important result was also that diversification did not serve as an immunity to risk. During volatile periods, the diversified portfolio also spiked dramatically in volatility, when all the assets started moving in unison, making the diversification fail. This was further illustrated by studying the pairwise correlation structure. In low correlation regimes, the effectiveness of diversification was high; however, as the correlation and volatility increased during moments of financial instability, the diversification benefit quickly started to decline.

Ultimately, the study showed that while diversification with the found asset size of 25 stocks served as a good protection against the unsystematic risk, it did not protect against market-wide movements. An important conclusion from the study was therefore that diversification works as a good tool for eliminating unsystematic risk during normal market conditions, but ultimately fails during systemic crises.

Finally, it is also important to consider that, as with any statistical model, the results are prone to model uncertainty. While the threshold found should not be interpreted as a universal constant, it still worked as a valuable tool to illustrate the benefits of diversification. For the purpose of analyzing portfolio volatility and capturing the heavy-tailed nature of the data, the model served as a sophisticated approximation of the diversification effect. It is, however, also important to note that the study mainly focused on the risk, not the return. By removing the uncertainty and adding more assets to the portfolio, the possibility of higher returns is also averaged out. By diversifying, the investor is protected against the big losers, but could also miss out on the big winners. It therefore ultimately becomes a trade-off between risk and return.

## References

- Bodie, Z., A. Kane, and A. J. Marcus (2013). *Essentials of Investments*. 9th ed. New York: McGraw-Hill/Irwin.
- Genz, Alan and Frank Bretz (2009). *mvtnorm package*. Lecture Notes in Statistics. Heidelberg: Springer-Verlag. ISBN: 978-3-642-01688-2. URL: <https://cran.r-project.org/web/packages/mvtnorm/index.html>.
- Glasserman, Paul (2003). *Monte Carlo Methods in Financial Engineering*. 1st. Vol. 53. New York, NY: Springer New York. URL: <https://doi.org/10.1007/978-0-387-21617-1>.
- Harvey, Campbell R. et al. (2010). “Portfolio Selection with Higher Moments”. In: *Quantitative Finance* 10.5, pp. 469–485. DOI: <https://doi.org/10.1080/14697681003756877>.
- Kahneman, Daniel and Amos Tversky (1979). “Prospect Theory: An Analysis of Decision under Risk”. In: *Econometrica* 47.2, pp. 263–291. URL: <https://doi.org/10.2307/1914185>.
- Liu, Chuanhai and Donald B. Rubin (1995). “ML Estimation of the t Distribution Using EM and Its Extensions, ECM and ECME”. In: *Statistica Sinica* 5.1, pp. 19–39. URL: <http://www.jstor.org/stable/24305551>.
- Mandelbrot, Benoit (Oct. 1967). “The Variation of Some Other Speculative Prices”. In: *The Journal of Business* 40.4, pp. 393–413. URL: <https://www.jstor.org/stable/2351623>.
- Markowitz, Harry (1952). “Portfolio Selection”. In: *The Journal of Finance* 7.1, pp. 77–91. URL: <https://doi.org/10.2307/2975974>.
- Palomar, D. P. et al. (2023). *fitHeavyTail: Mean and Covariance Matrix Estimation under Heavy Tails*. R package version 0.2.0. URL: <https://CRAN.R-project.org/package=fitHeavyTail>.